

1-2

publish

4

rely

3, 5-7

maintain and improve



no unique repair

**1 Learn**

analysis optimizer, an LLM agent

question: March store revenue  
`SUM(ss_net_paid) ... d_moy = 3`  
 13,570,368.70, judged correct ✓  
 answer provenance extracted

**2 Publish revision 0**

after the admission checks

key uniqueness: one row per sale ✓  
 date role: d\_date\_sk unique ✓  
 join completeness: loss ≤ 1.13% ✓  
 evidence: learning SQL, snapshot

**3 Optimize: revision 1**

period as a date-key range

equal on 14 periods ✓  
 29.1 → 23.3 ms, -20% ✓  
 adds date-key contiguity  
 revision 0 stays usable

**4 User agents rely on it**

given only the metric name

September store revenue?  
 lookup: revision 1, valid ✓  
 declared, run on one snapshot ✓  
 13,069,651.80, correct ✓

a later write

**5 Restatement**

old sales kept as non-current rows

store\_sales 13 → 14  
 key uniqueness fails ✗  
 revision 1 invalid ✗  
 September: 14,300,525.64 ✗

**6 Bounded repair**

filters on low-cardinality columns

ss\_is\_current='1' ✓  
 ss\_is\_current='0' ✗  
 exactly one restores the grain ✓

**7 Publish revision 2**

regression test as of learning time

March, learning SQL 13,570,368.70  
 March, repaired SQL 13,570,368.70  
 equal: repair accepted ✓  
 September: 12,932,888.04 ✓

**✗ No unique repair**

e.g., a duplicate load

no filter restores the grain ✗  
 or several filters disagree  
 definition invalidated  
 relearned from new tasks

none or several